# Write-up

An analyst co-pilot that clarifies, checks definitions, picks the right data layer and shows its evidence.

## 01 What I built and why

**1 Clarify** — metric, period, population; asks if ambiguous

**2 Verify definition** — metric exists and its definition in registry + catalog

**3 Route to a layer** — Gold if it fits; Silver for detail; Bronze for raw audit

**4 Query + evidence** — read-only, bounded; one log per query, never overwritten

**5 Test hypotheses** — ≤3 per round, ≤3 rounds; stop when explained

**6 Report** — pattern ≠ cause; fixed PDF template, ≤10 pages

**Read by every run, in Claude Code or Codex:** AGENTS.md · SKILL.md · catalog · registry

Guardrails: read-only authorizer · row and time limits

**Gold** — 8 monthly tables · fast standard answers

**Silver** — 10 cleaned tables · detail for investigations

**Bronze** — 20 views over raw tables · audits

*One SQLite database; layers built in one transaction with 122 reconciliation checks. Step 3 decides which layer each query reads.*

- **Clarifies instead of assuming.** When the metric, period or meaning is open, it asks first and states every assumption it keeps.
- **Verifies the metric before computing it.** It confirms the metric exists and uses its approved definition and SQL from the metric registry, plus joins and known gaps from the data catalog. Unsupported requests are declined, not improvised.
- **Maps the question to a layer for speed.** Gold answers standard monthly questions directly; Silver keeps per-user detail; Bronze is only for auditing raw values. Fewer, smaller queries keep answers fast.
- **Evidence for every number and LLM-neutral by design.** Each query is logged with its SQL, layer and metric version. Definitions, rules and guardrails live in files, not prompts, so Claude Code and Codex behave the same and a lesson becomes a rule for every later run.

## 02 How it performed: 18 blind test questions, two rounds

| | Q01 | Q02 | Q03 | Q04 | Q05 | Q06 | Q07 | Q08 | Q09 | Q10 | Q11 | Q12 | Q13 | Q14 | Q15 | Q16 | Q17 | Q18 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Round 1** | Pass | Fail | Pass | Pass | Pass | Pass | Fail | Pass | Pass | Pass | Partial | Pass | Pass | Pass | Pass | Pass | Pass | Pass |
| **Round 2** | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Pass | Partial | Pass | Pass | Pass | Pass | Pass | Pass | Pass |

**Round 1: 15 pass · 1 partial · 2 fail** **Round 2: 17 · 1 · 0**

| Question type | Questions | Result | What happened |
|---|---|---|---|
| **Simple numbers and filters** | Q01 Q05 Q08 Q17 Q18 | All pass | Exact to the user and the cent |
| **Vague questions** | Q03 Q10 Q15 Q18 | All pass | Asked, or stated a clear choice, before picking a metric, period or meaning |
| **Unanswerable question** | Q16 | Pass | Declined and named the data that's missing |
| **Data-quality traps** | Q12 Q13 Q14 | All pass | Caught both planted issues; didn't claim to have detected a reviewed one |
| **Comparisons and lifecycle** | Q04 Q06 Q09 | All pass | Right time basis; groups reconcile to the total |
| **Cohort conversion** | Q07 | Fail → Pass | Treated April, the start-up month, as a normal baseline; a new skill rule fixed it |
| **Funnel drop-off** | Q11 | Partial | Right answer, but read April's start-up as a funnel decline; the rule didn't trigger |
| **GM investigation** | Q02 | Fail → Pass | v1 right on numbers but too firm on some conclusions; after three certainty rules, v2 and a blind rerun both passed |

Scored against expected values computed from the raw tables, not the agent's layers. Correct numbers alone don't pass: a silent choice of definition, a missing caveat or an overstated conclusion fails. Round 2: Q02 was revised after review and rerun blind; Q07 and Q11 were rerun, with Q01 and Q09 to check nothing broke; other results carry over. **Hardest:** comparisons that need context the data doesn't announce, such as a start-up month.

## 03 Running it for a 20-person product team

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|---|---|---|---|
| **Time to answer** (observed) | **Simple** (MAU by month): 2–3 min | **Normal** (refund impact): 5–6 min | **Complex** (MAU up, revenue flat): 10–12 min |

- **How the team uses it:** analysts and PMs ask in plain English from Claude Code or Codex (later Slack or a web page), all on one shared repository, so definitions and guardrails are the same for every person and model.
- **Set-up:** build a data catalog and metric registry for the team's warehouse, with the analytics lead approving each definition, then the Gold and Silver layers the agent routes to.
- **Cost:** mostly model tokens; queries are nearly free. Not measured here: log tokens per question, then budget by question type. Answering from the smallest layer that fits keeps both cost and latency down.
- **Guardrails:** read-only access (database role plus the query tool's check), row and time limits, an evidence log behind every figure, no sensitive columns in the catalog, and a question back when the ask is unclear.
- **Freshness:** real data keeps arriving, unlike this dataset; the agent first checks a freshness tracker (last load per source and layer) and warns or stops when data is stale.
- **Quality over time:** a test set of questions with known answers (these 18, growing from real ones), rerun on every change to the skill, definitions or data; track pass rate, clarification rate, query errors and user ratings; turn failures into rules.

## 04 With more time

- Run the same questions in both runtimes for a direct Claude vs Codex comparison, plus held-out questions.
- Measure tokens and time per question to put a real cost on each question type.
- A helper that drafts the catalog and metric registry for a new database; a Slack or web front end.

**Tools:** Claude Code (Claude Opus 5.5, medium) built the framework, ran the GM investigation and the answer key; Codex (GPT Terra 6.1, medium) ran the evaluation; ChatGPT (Terra 5.6, later 6.1, medium) reviewed independently.